# UrjaKhata system architecture

One metered ledger for an MSME foundry: three meters in, three proofs out (owner savings, buyer carbon passport, lender savings report)

BUILT working in the prototype (simulated plant) SIM simulated by the digital twin in the prototype NEXT in the pilot roadmap

## 1. Plant floor

three meters, no wiring into machines

**M1 mains meter, Class 1 Modbus** SIM
e.g. Schneider EasyLogic PM2220; CT 1000/5 A

**M2 furnace meter** SIM
415 V, 50 Hz side of the SCR/IGBT converter

**M3 compressor meter** SIM
load, unload and night running

**Heat log** SIM
charge weight and mix, pyrometer, slag, rejects (WhatsApp or voice)

**LPG bills, pig iron certificates** NEXT
read by OCR

**Shielded RS485 bus** NEXT
120 ohm termination, surge protection

## 2. Edge gateway

4G or Wi-Fi, low cost

**Modbus RTU polling** NEXT
power 1 s, energy 15 s, PF and THD 60 s

**7-day local buffer** NEXT
survives network outages

**Data completeness check** NEXT
CRC errors and gaps reported

**Encrypted MQTT** NEXT
to the cloud ledger

**Foundry digital twin** BUILT
BEE-calibrated physics; stands in for plant and gateway

## 3. Cloud ledger (Python)

one record per heat, per day, per shipment

**Hybrid NILM** BUILT
gradient boosting plus physics rules; confidence tier per machine

**SEC engine** BUILT
kWh per liquid tonne and per good tonne, every heat

**Drift detector** BUILT
CUSUM on residuals of a practice regression

**ToD melt optimiser** BUILT
MILP (HiGHS) on real PGVCL, MSEDCL, TNPDCL tariffs; shift limits

**Carbon engine** BUILT
CBAM boundary (direct + pig iron) and Scope 1+2 (CEA 0.675)

**M&V engine** BUILT
IPMVP Option B and C with ASHRAE 14 uncertainty

## 4. Outputs

who uses each one

**Owner: daily briefing** BUILT
top 3 actions in rupees; English and Hindi; WhatsApp in pilot

**Dashboard with guided demo** BUILT
static site on Vercel

**EU buyer: carbon passport** BUILT
CBAM boundary per shipment, evidence trail, JSON

**Lender: savings proof** BUILT
for ADEETIE, SIDBI 4E, ESCOs

**ERP or Tally sync** NEXT
production and dispatch

**Energy data**
Meters to gateway to ledger: kW every second, kWh every 15 s, stored 7 days at the edge, aggregated per heat and per day

**Production and fuel data**
Operator log, WhatsApp or voice entries, OCR of bills, ERP; links energy to good tonnes

**Money flow**
Owner saves on the bill and pays the subscription; lenders and ESCOs pay per verified project; buyers keep orders